SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1701**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 **Total Marks:30**

Annexure A **Dry**

Run Evaluation

Code of Conduct:

I JAYA PRAKASH M / 1912260496 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

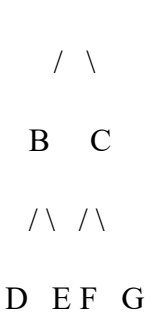
Signature of the student: JAYA PRAKASH M

Annexure A

Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Consider the following graph:



Task: Starting from node **A**, perform a **Breadth-First Search (BFS)**. Complete the following table.

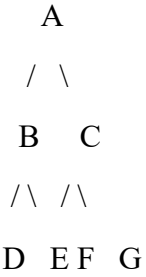
Iteration Queue Visited Node

1
2
3
4
5
6
7

Questions:

1. Write the BFS traversal order.
2. Show the queue contents after each iteration.

2. Depth-First Search (DFS) Using the same graph,



Iteration	Stack	Visited Node
1		
2		
3		
4		
5		
6		
7		

Perform **Depth-First Search (DFS)** starting from **A**.

Complete the table.

3. **Initial State :**
   1 3 6
   5 0 2
   4 7 8

Goal State :
1 2 3
4 5 6
7 8 0

Task

Trace **Breadth-First Search (BFS)** until the goal state is reached.

Fill in the table.

Iteration Current State Queue Before Expansion Queue After Expansion
1
2
3
4
5

Questions

1. Which node is expanded in each iteration?
2. How many states are generated in total?
3. Write the complete solution path from the initial state to the goal state.
4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	**20**	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	**30**	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.
Application of Search Algorithm	**20**	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	**20**	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Presentation and Logical Reasoning	**10**	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

ANNEXURE A – DRY RUN EVALUATION

1. Question 1 – Breadth-First Search (BFS)

Given graph (left child is visited before right child):

A\n / \\\n B C\n / \\ / \\\n D E F G

BFS visits nodes level by level, from left to right.

Iteration	Queue Before Processing	Visited Node	Queue After Processing
1	A	A	B, C
2	B, C	B	C, D, E
3	C, D, E	C	D, E, F, G
4	D, E, F, G	D	E, F, G
5	E, F, G	E	F, G
6	F, G	F	G
7	G	G	Empty

1. BFS traversal order: A → B → C → D → E → F → G

2. Queue contents after each iteration: (1) B, C; (2) C, D, E; (3) D, E, F, G; (4) E, F, G; (5) F, G; (6) G; (7) Empty.

2. Question 2 – Depth-First Search (DFS)

DFS explores one branch as deeply as possible before backtracking. The left child is visited first.

Iteration	Visited Node	Active Path (Stack Concept)
1	A	A
2	B	A, B
3	D	A, B, D
4	E	A, B, E
5	C	A, C
6	F	A, C, F
7	G	A, C, G

DFS traversal order: A → B → D → E → C → F → G

Note: The stack column shows the active path conceptually; an iterative DFS stack may differ depending on when children are pushed and popped.

3. Question 3 – 8-Puzzle Using Breadth-First Search (BFS)

Initial State:

1 3 6\n5 0 2\n4 7 8

Goal State:

1 2 3\n4 5 6\n7 8 0

Here, 0 represents the empty space. For compact state notation, "_" represents the empty space and "/" separates rows.

A. BFS dry-run table (first five iterations)

Iteration	Current State (Expanded)	Queue Before Expansion	Queue After Expansion
1	136/5_2/478	136/5_2/478	1_6/532/478; 136/_52/478; 136/52_/478; 136/572/4_8
2	1_6/532/478	1_6/532/478; 136/_52/478; 136/52_/478; 136/572/4_8	136/_52/478; 136/52_/478; 136/572/4_8; _16/532/478; 16_/532/478
3	136/_52/478	136/_52/478; 136/52_/478; 136/572/4_8; _16/532/478; 16_/532/478	136/52_/478; 136/572/4_8; _16/532/478; 16_/532/478; _36/152/478; 136/452/_78
4	136/52_/478	136/52_/478; 136/572/4_8; _16/532/478; 16_/532/478; _36/152/478; 136/452/_78	136/572/4_8; _16/532/478; 16_/532/478; _36/152/478; 136/452/_78; 13_/526/478; 136/528/47_
5	136/572/4_8	136/572/4_8; _16/532/478; 16_/532/478; _36/152/478; 136/452/_78; 13_/526/478; 136/528/47_	_16/532/478; 16_/532/478; _36/152/478; 136/452/_78; 13_/526/478; 136/528/47_; 136/572/_48; 136/572/48_

B. Questions and Answers

1. Which node is expanded in each iteration?

The first five expanded states are:

- Iteration 1: 136/5_2/478
- Iteration 2: 1_6/532/478
- Iteration 3: 136/_52/478
- Iteration 4: 136/52_/478

- Iteration 5: 136/572/4_8

2. How many states are generated in total?

Using BFS with duplicate-state checking and the move order used in this trace:

- Distinct states discovered by the time the goal is reached: 456 (including the initial and goal states).
- States removed from the queue up to and including the goal: 283.
- States in the solution path: 9 (including the initial and goal states).

The total can differ depending on the definition of "generated" and the implementation.

3. Complete solution path from the initial state to the goal state:

Initial	1 3 6\n5 0 2\n4 7 8
Move 1	1 3 6\n5 2 0\n4 7 8
Move 2	1 3 0\n5 2 6\n4 7 8
Move 3	1 0 3\n5 2 6\n4 7 8
Move 4	1 2 3\n5 0 6\n4 7 8
Move 5	1 2 3\n0 5 6\n4 7 8
Move 6	1 2 3\n4 5 6\n0 7 8
Move 7	1 2 3\n4 5 6\n7 0 8
Goal – Move 8	1 2 3\n4 5 6\n7 8 0

The solution path requires 8 moves.

4. Why does BFS always find the shortest solution in an unweighted 8-puzzle problem?

BFS explores states level by level. It first explores states requiring one move, then two moves, then three moves, and so on. Since every move has equal cost, the first time BFS reaches the goal, the solution has the minimum number of moves.